Spark DataFrame Case Study

E-Commerce Analytics Platform

Overview

A large e-commerce company wants to analyze its sales performance, customer behavior, product profitability, and return trends. The organization stores transactional data across multiple datasets and requires an analytics solution built using Apache Spark DataFrames.

You have been provided with multiple datasets containing customer, product, order, order item, and return information. Your task is to build analytical outputs that will help business leaders make data-driven decisions. The datasets contain millions of records and should be processed efficiently using Spark. The final deliverables should include analytical outputs and an exported business-ready dataset.

Dataset Information

Dataset	Approx. Volume	Schema Overview
customers.csv	~100,000 records	`customer_id`, `customer_name`, `city`, `state`, `registration_da` `customer_segment`
products.csv	~50,000 records	`product_id`, `product_name`, `category`, `brand`, `unit_cost`
orders.csv	~1,000,000 records	`order_id`, `customer_id`, `order_date`, `payment_mode`, `order_s`
order_items.csv	~3,000,000 records	`order_item_id`, `order_id`, `product_id`, `quantity`, `selling_p`
returns.csv	~100,000 records	`return_id`, `order_id`, `return_date`, `return_reason`

Analytical Questions

1. Calculate the total number of customers, products, orders, and returned orders.
2. Find the total sales amount generated by each product category.
3. Find the Top 10 customers based on total purchase amount.
4. Calculate monthly sales trends for the last available year.
5. Find the return percentage for each product category.
6. Determine the most preferred payment mode in each state.
7. Identify customers who have purchased products from at least 5 different categories and spent more than 1,00,000.
8. Find the Top 3 products by revenue within each category.

9. Calculate customer lifetime value (CLV) and classify customers into Bronze, Silver, and Gold segments.
10. Create a final business analytics dataset containing customer, order, revenue, return, and segmentation information and export it in Parquet format partitioned by state.